CS4200 Final Project – United States Gas Prices from Jan 2026 – Mar 2026

Overview

This project was for my Big Data Analysis class. In this project I explore and analyze a data set on gas and oil prices since before the Iran US conflict in 2026 until end of march 2026.

The Data Set

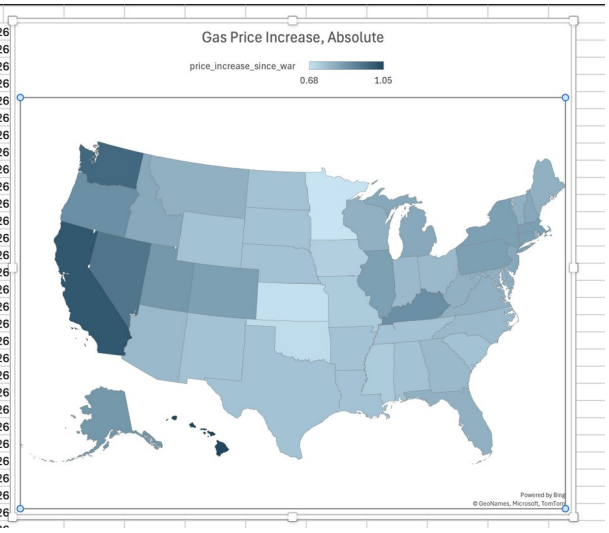
The data set that was used for this project was the Iran War Oil Shock 2026 | Brent & Gas Tracker. This data set was found on kaggle with a gold rating and 10.0 usability. The GetDataSet.py file in my project is a script to grab the data set from kaggle and place it into the users working directory.

```python
import kagglehub

path = kagglehub.dataset_download(
        "alitaqishah/iran-war-oil-shock-2026-brent-and-gas-tracker",
        output_dir="./data")

print("Path to dataset files:", path)
```

Data Exploration

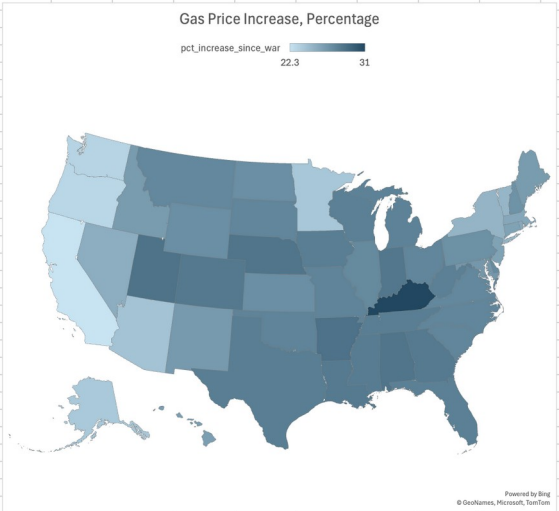
The data set is split into three different CSV files iran_war_gas_prices_by_state.csv, iran_war_key_events_timeline.csv, and iran_war_oil_prices_daily_2026.csv. In this project I explored the iran_war_gas_prices_by_state.csv and iran_war_oil_prices_daily_2026.csv in excel. At first I was interested in evaluating the different US. states and how they were effected by the change in gas prices. To show some of the effects I sorted and filtered the data and created some visualizations. The first visualization is the states absolute price increase from Jan 8 to Mar 19.

state	region	gas_price_r	gas_price_p	gas_price_k	price_increa	pct_increas	price_vs_na	source
Hawaii	West	5.07	4.02	3.78	1.05	26.1	1.19	AAA State Gas Price Averages â€" March 19, 2026
California	West	5.53	4.52	3.88	1.01	22.3	1.65	AAA State Gas Price Averages â€" March 19, 2026
Washingtor	West	5.15	4.18	3.42	0.97	23.2	1.27	AAA State Gas Price Averages â€" March 19, 2026
Nevada	West	4.66	3.72	3.18	0.94	25.3	0.78	AAA State Gas Price Averages â€" March 19, 2026
Kentucky	South	3.72	2.84	2.48	0.88	31	-0.16	AAA State Gas Price Averages â€" March 19, 2026
Oregon	West	4.7	3.82	3.22	0.88	23	0.82	AAA State Gas Price Averages â€" March 19, 2026
Utah	West	3.88	3.02	2.72	0.86	28.5	0	AAA State Gas Price Averages â€" March 19, 2026
Alaska	West	4.48	3.62	3.12	0.86	23.8	0.6	AAA State Gas Price Averages â€" March 19, 2026
Colorado	West	3.82	2.98	2.68	0.84	28.2	-0.06	AAA State Gas Price Averages â€" March 19, 2026
Illinois	Midwest	3.92	3.08	2.72	0.84	27.3	0.04	AAA State Gas Price Averages â€" March 19, 2026
Pennsylvar	Northeast	3.96	3.12	2.74	0.84	26.9	0.08	AAA State Gas Price Averages â€" March 19, 2026
Connecticu	Northeast	4.08	3.24	2.8	0.84	25.9	0.2	AAA State Gas Price Averages â€" March 19, 2026
Massachus	Northeast	4.12	3.28	2.84	0.84	25.6	0.24	AAA State Gas Price Averages â€" March 19, 2026
New York	Northeast	4.22	3.38	2.92	0.84	24.9	0.34	AAA State Gas Price Averages â€" March 19, 2026
Michigan	Midwest	3.78	2.96	2.64	0.82	27.7	-0.1	AAA State Gas Price Averages â€" March 19, 2026
New Hamps	Northeast	3.88	3.06	2.68	0.82	26.8	0	AAA State Gas Price Averages â€" March 19, 2026
Idaho	West	3.94	3.12	2.74	0.82	26.3	0.06	AAA State Gas Price Averages â€" March 19, 2026
New Jersey	Northeast	3.98	3.16	2.76	0.82	25.9	0.1	AAA State Gas Price Averages â€" March 19, 2026
Wisconsin	Midwest	3.68	2.88	2.58	0.8	27.8	-0.2	AAA State Gas Price Averages â€" March 19, 2026
Florida	South	3.68	2.88	2.58	0.8	27.8	-0.2	AAA State Gas Price Averages â€" March 19, 2026
West Virgini	South	3.68	2.88	2.54	0.8	27.8	-0.2	AAA State Gas Price Averages â€" March 19, 2026
Montana	West	3.72	2.92	2.62	0.8	27.4	-0.16	AAA State Gas Price Averages â€" March 19, 2026
Virginia	South	3.72	2.92	2.58	0.8	27.4	-0.16	AAA State Gas Price Averages â€" March 19, 2026
Delaware	Northeast	3.74	2.94	2.58	0.8	27.2	-0.14	AAA State Gas Price Averages â€" March 19, 2026
Maryland	Northeast	3.82	3.02	2.64	0.8	26.5	-0.06	AAA State Gas Price Averages â€" March 19, 2026
Maine	Northeast	3.84	3.04	2.66	0.8	26.3	-0.04	AAA State Gas Price Averages â€" March 19, 2026
Rhode Islar	Northeast	3.94	3.14	2.76	0.8	25.5	0.06	AAA State Gas Price Averages â€" March 19, 2026
Vermont	Northeast	4.02	3.22	2.78	0.8	24.8	0.14	AAA State Gas Price Averages â€" March 19, 2026
Indiana	Midwest	3.54	2.76	2.48	0.78	28.3	-0.34	AAA State Gas Price Averages â€" March 19, 2026



As you can see from this graph the range of absolute increase goes from light to dark, meaning the lighter colored states had less of an absolute increase while the darker colored states had more of an absolute increase. This second graph shows the percentage increase from light to dark.

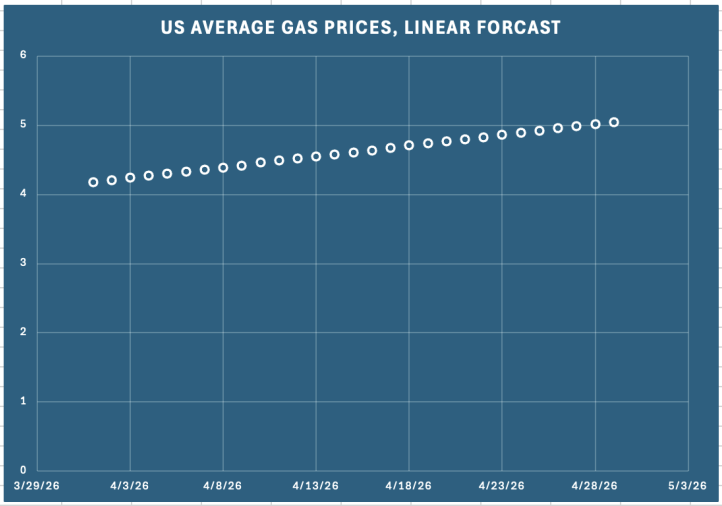
state	region	gas_price_r	gas_price_p	gas_price_ja	price_increa	pct_increas	price_vs_na	source
Kentucky	South	3.72	2.84	2.48	0.88	31	-0.16	AAA State Gas Price Averages â€" March 19, 2026
Arkansas	South	3.42	2.66	2.36	0.76	28.6	-0.46	AAA State Gas Price Averages â€" March 19, 2026
Utah	West	3.88	3.02	2.72	0.86	28.5	0	AAA State Gas Price Averages â€" March 19, 2026
Nebraska	Midwest	3.44	2.68	2.38	0.76	28.4	-0.44	AAA State Gas Price Averages â€" March 19, 2026
Alabama	South	3.44	2.68	2.38	0.76	28.4	-0.44	AAA State Gas Price Averages â€" March 19, 2026
Indiana	Midwest	3.54	2.76	2.48	0.78	28.3	-0.34	AAA State Gas Price Averages â€" March 19, 2026
Colorado	West	3.82	2.98	2.68	0.84	28.2	-0.06	AAA State Gas Price Averages â€" March 19, 2026
Georgia	South	3.56	2.78	2.48	0.78	28.1	-0.32	AAA State Gas Price Averages â€" March 19, 2026
Louisiana	South	3.46	2.7	2.4	0.76	28.1	-0.42	AAA State Gas Price Averages â€" March 19, 2026
Mississippi	South	3.38	2.64	2.34	0.74	28	-0.5	AAA State Gas Price Averages â€" March 19, 2026
Iowa	Midwest	3.35	2.62	2.34	0.73	27.9	-0.53	AAA State Gas Price Averages â€" March 19, 2026
Tennessee	South	3.48	2.72	2.42	0.76	27.9	-0.4	AAA State Gas Price Averages â€" March 19, 2026
Texas	South	3.48	2.72	2.44	0.76	27.9	-0.4	AAA State Gas Price Averages â€" March 19, 2026
Wisconsin	Midwest	3.68	2.88	2.58	0.8	27.8	-0.2	AAA State Gas Price Averages â€" March 19, 2026
Florida	South	3.68	2.88	2.58	0.8	27.8	-0.2	AAA State Gas Price Averages â€" March 19, 2026
West Virgini	South	3.68	2.88	2.54	0.8	27.8	-0.2	AAA State Gas Price Averages â€" March 19, 2026
Michigan	Midwest	3.78	2.96	2.64	0.82	27.7	-0.1	AAA State Gas Price Averages â€" March 19, 2026
Missouri	Midwest	3.42	2.68	2.38	0.74	27.6	-0.46	AAA State Gas Price Averages â€" March 19, 2026
Oklahoma	South	3.24	2.54	2.28	0.7	27.6	-0.64	AAA State Gas Price Averages â€" March 19, 2026
Ohio	Midwest	3.62	2.84	2.54	0.78	27.5	-0.26	AAA State Gas Price Averages â€" March 19, 2026
South Dakc	Midwest	3.52	2.76	2.46	0.76	27.5	-0.36	AAA State Gas Price Averages â€" March 19, 2026
North Carol	South	3.62	2.84	2.54	0.78	27.5	-0.26	AAA State Gas Price Averages â€" March 19, 2026
South Caro	South	3.52	2.76	2.46	0.76	27.5	-0.36	AAA State Gas Price Averages â€" March 19, 2026
Montana	West	3.72	2.92	2.62	0.8	27.4	-0.16	AAA State Gas Price Averages â€" March 19, 2026
Virginia	South	3.72	2.92	2.58	0.8	27.4	-0.16	AAA State Gas Price Averages â€" March 19, 2026
Illinois	Midwest	3.92	3.08	2.72	0.84	27.3	0.04	AAA State Gas Price Averages â€" March 19, 2026
Delaware	Northeast	3.74	2.94	2.58	0.8	27.2	-0.14	AAA State Gas Price Averages â€" March 19, 2026
Wyoming	West	3.58	2.82	2.52	0.76	27	-0.3	AAA State Gas Price Averages â€" March 19, 2026
North Dako	Midwest	3.58	2.82	2.52	0.76	27	-0.3	AAA State Gas Price Averages â€" March 19, 2026



From these two graphs we can see that while the absolute increase affected most heavily the western states, the percentage increase affected the Midwest / southern states the most.

After a few more graphs and sorts I noticed in the iran_war_oil_prices_daily_2026.csv that the data set had a US. Average gas price column. Noticing this I was excited to do some predictive analysis on this data set and in excel charted out a linear forecast to see what the gas prices could have potentially been in early April. It was not accurate.

date	us_gas_avg_gallon		Future Dates	us_gas_avg_gallon Forcast
2/9/26	2.81		4/1/26	4.18
2/10/26	2.83		4/2/26	4.21
2/14/26	2.84		4/3/26	4.24
2/18/26	2.85		4/4/26	4.27
2/24/26	2.87		4/5/26	4.3
2/27/26	2.92		4/6/26	4.33
2/28/26	2.98		4/7/26	4.36
3/1/26	3.05		4/8/26	4.39
3/2/26	3.12		4/9/26	4.42
3/3/26	3.18		4/10/26	4.46
3/4/26	3.26		4/11/26	4.49
3/5/26	3.27		4/12/26	4.52
3/7/26	3.36		4/13/26	4.55
3/10/26	3.48		4/14/26	4.58
3/11/26	3.54		4/15/26	4.61
3/12/26	3.58		4/16/26	4.64
3/13/26	3.63		4/17/26	4.67
3/14/26	3.63		4/18/26	4.71
3/15/26	3.65		4/19/26	4.74
3/16/26	3.72		4/20/26	4.77
3/17/26	3.79		4/21/26	4.8
3/18/26	3.84		4/22/26	4.83
3/19/26	3.88		4/23/26	4.86
3/20/26	3.92		4/24/26	4.89
			4/25/26	4.92
			4/26/26	4.96
			4/27/26	4.99
			4/28/26	5.02
			4/29/26	5.05



Data Wrangling

I decided that I wanted to do some more predictive analysis on the Average US gas price and see what values effected it and what not. Before doing any analysis I decided it would be good to cleanse the data. To preform some of this data wrangling I used python with pandas and sklearns DBSCAN, Oh and also the book.

In python I imported the csv checked the first few rows, checked if any files were duplicate, and checked if any values were null. Lastly I created a DBSCAN to check for any outliers on the gas price. Below I have the output.

```
(24, 16)
         date  brent_usd_barrel  ...                                          source      data_as_of
0  2026-02-09              67.4  ...  AAA/CNBC/Fortune/NPR/CBS/Reuters/Euronews veri...  March 20, 2026
1  2026-02-10              69.3  ...  AAA/CNBC/Fortune/NPR/CBS/Reuters/Euronews veri...  March 20, 2026
2  2026-02-14              71.2  ...  AAA/CNBC/Fortune/NPR/CBS/Reuters/Euronews veri...  March 20, 2026
3  2026-02-18              72.8  ...  AAA/CNBC/Fortune/NPR/CBS/Reuters/Euronews veri...  March 20, 2026
4  2026-02-24              71.5  ...  AAA/CNBC/Fortune/NPR/CBS/Reuters/Euronews veri...  March 20, 2026

[5 rows x 16 columns]
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18    False
19    False
20    False
21    False
22    False
23    False
dtype: bool
date                            0
brent_usd_barrel                0
wti_usd_barrel                  0
dubai_usd_barrel                0
us_gas_avg_gallon               0
us_diesel_avg_gallon            0
strait_hormuz_daily_ships       0
iran_production_mbpd            0
key_event                       0
war_day                         0
brent_vs_prewar_pct             0
gas_vs_prewar_pct               0
gas_change_from_prewar_dollars  0
phase                           0
source                          0
data_as_of                      0
dtype: int64
Index   Cluster         X       Y
```

As you can see from this output the data was pretty clean.

Predictive analysis

So after cleansing and exploring the data I was really excited to see how the different variables that were given in the data set could effect the average US. Gas price. I live in the United States and so understanding how these variables effected our gas prices was very interesting. To preform this predictive analysis I used the Random Forest Regressor from sklearn, numpy, pandas, and python. The variables that were assigned to the x value were brent_usd_barrel, wti_usd_barrel, dubai_usd_barrel, strait_hormuz_daily_ships, iran_production_mbpd, war_day, brent_vs_prewar_pct, gas_vs_prewar_pct, and gas_change_from_prewar_dollars. The Y value was assigned to ‘us_gas_avg_gallon. Below is the output from running the prediction.

```
[0.10713779 0.09834093 0.11899949 0.07515496 0.14043444 0.14063492
 0.11754452 0.07089984 0.13085311]
Actual:  2.81 Predicted:  2.818500000000002
Actual:  2.83 Predicted:  2.8302000000000023
Actual:  2.84 Predicted:  2.8451999999999993
Actual:  2.85 Predicted:  2.8567999999999993
Actual:  2.87 Predicted:  2.873400000000002
Actual:  2.92 Predicted:  2.9078999999999997
Actual:  2.98 Predicted:  2.990000000000001
Actual:  3.05 Predicted:  3.064000000000005
Actual:  3.12 Predicted:  3.1249000000000042
Actual:  3.18 Predicted:  3.195600000000004
Actual:  3.26 Predicted:  3.258799999999998
Actual:  3.27 Predicted:  3.2661000000000002
Actual:  3.36 Predicted:  3.3710000000000058
Actual:  3.48 Predicted:  3.502300000000003
Actual:  3.54 Predicted:  3.5430000000000046
Actual:  3.58 Predicted:  3.5904000000000003
Actual:  3.63 Predicted:  3.6449999999999982
Actual:  3.63 Predicted:  3.631199999999997
Actual:  3.65 Predicted:  3.6326999999999985
Actual:  3.72 Predicted:  3.6919000000000026
Actual:  3.79 Predicted:  3.769500000000005
Actual:  3.84 Predicted:  3.8458999999999945
Actual:  3.88 Predicted:  3.869599999999997
Actual:  3.92 Predicted:  3.8657999999999992
```

From this output we can see that with all the other factors that I included the model was pretty accurately predicting what the average gas price was going to be in the US.

Conclusion.

In the future I would like to dig deeper into the predictive analysis aspect of this project. I would run some other test to find out which variables were most likely to predict the gas prices and with that information make some more predictive models for the data set. There is also a key events csv in this data set that deals with key events from the war. In the future I would like to incorporate this knowledge to help with the predictions to see what sort of events had the greatest effect on the gas prices. But for what I have done with this project with the knowledge I have I feel satisfied with the outcomes.